

Akash Sarkar

M.Tech in Cyber Forensics & Information Security
College of Engineering Kallooppara
Portfolio Website

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EXPERIENCE

Ethical Hacking & Penetration Testing Intern

Jun 2025 – Jul 2025

C-DAC

Remote

- Executed network and web-based VAPT on enterprise infrastructures.
- Identified 30+ vulnerabilities using Nmap, Metasploit, and Wireshark.
- Authored OWASP-aligned remediation reports, enabling the patching of critical vulnerabilities and significantly reducing risk exposure.

Cybersecurity & Digital Forensics Intern

Apr 2025 – Jun 2025

Academy of Skill Development

Kolkata, IN

- Conducted VAPT and in-depth forensic analysis using tools like Volatility and Autopsy on compromised test environments.
- Assessed attack patterns to implement system hardening and risk reduction.

Cybersecurity Internship (Python Focus)

Dec 2022 – Jan 2023

Ardent Computech Pvt. Ltd

Kolkata, IN

- Developed Python-based tools for network scanning and log analysis.
- Simulated exploitation scenarios in Linux to analyze system weaknesses.

PROJECTS

ThreatWeaver: Zero Trust Security Engine

Python, JavaScript, Shell, Active Directory, Machine Learning, RAG

- Built an AI-driven threat detection pipeline that transforms Active Directory logs into structured threat intelligence using Isolation Forest anomaly detection and Retrieval-Augmented Generation (RAG)-based LLM analysis.
- Integrated MITRE ATT&CK mapping for standardized adversarial technique classification and evidence-based security reporting.
- Developed automated log ingestion, enrichment, and summarization modules to accelerate SOC analyst triage and incident investigation workflows.

Secure File Storage System

Flask, MongoDB, AES-256, RBAC

- Deployed a secure file storage server on a Linux VPS using Nginx, featuring AES encryption and 2FA.
- Developed blockchain-based audit logs to ensure record immutability.
- Integrated Role-Based Access Control (RBAC) and real-time security alerts.

TECHNICAL SKILLS

Tools & Infrastructure: Nmap, Burp Suite, Wireshark, Metasploit, Active Directory

Programming: Python, JavaScript, Shell, SQL

Environments: Kali Linux, Windows

Domains: VAPT, Digital Forensics, Web Security, Zero Trust, MITRE ATT&CK, OWASP Top 10

EDUCATION

M.Tech, Cyber Forensics & Information Security

Oct 2024 – Present

College of Engineering Kallooppara, IN

CGPA: 8.67 (In Progress)

B.Tech, Computer Science & Engineering

Oct 2020 – Jun 2024

School of Engineering and Technology, Dimapur, IN

CGPA: 7.42

ACTIVITIES & CERTIFICATIONS

Coordinator, Techaura 2023

Organized technical events for 100+ participants.

Participant, TryHackMe Advent of Cyber 2025

Completed 24 security challenges.

Certifications

Cybersecurity Job Simulations: PwC Switzerland & ANZ Australia.